ADR4AI: LLM-Based Architectural Pattern Selection and ADR Generation for AI-Enabled Systems

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Working draft: Tasks 1.1–1.3 only. The pattern-catalog description is pending human review. This document is not a complete paper or a submission-ready manuscript.

I. Methodology

A. Research Questions and Study Scope

We organize the study around three research questions (Table I). The computational protocol treats a user story and its acceptance criteria as a requirement case, with the system purpose providing context. It assesses each catalog pattern against each case independently. The assessment distinguishes applicability, non-applicability, and insufficient information; a positive classification requires a supporting user-story or acceptance-criterion reference. These classifications are model outputs, not independently established ground truth.

RQ1 characterizes the model's classification outcomes rather than measuring their correctness. RQ2 compares the patterns classified as applicable with a separate retrieval baseline. This baseline retrieves catalog candidates and reranks them to a final Top-5 without making an LLM-based applicability judgment. Overlap is therefore a descriptive relationship between two output sets, not evidence of either approach's precision, recall, or superiority.

ADR generation is a separate stage: each pair classified as applicable supplies the requirement case, selected pattern, and applicability evidence for one generated ADR. Generation completeness and stored provenance establish the availability and traceability of these records, not their architectural quality. RQ3 examines expert assessments of these selected pairs and their corresponding ADRs, rather than adding another automatic applicability classification.

The prepared questionnaire measures decision adequacy for the stated requirements (Q1), decision grounding (Q2), presentation of relevant consequences and trade-offs (Q3), perceived usefulness as a decision record (Q4), and pattern applicability (Q5). These are five separate expert-assessed dimensions; they are not a validated composite quality score. Optional comments provide context for the ratings. Neither implementation feasibility, cost savings, novelty, nor actual improvement in architectural practice is directly measured by these items.

Experts evaluate cases for a system they know, and only pairs selected by the LLM are presented. Consequently, RQ3 can characterize expert judgments of the presented recommendations but cannot identify missed applicable patterns or estimate recall over the full set of assessed pairs. Expert agreement with a recommendation does not by itself establish implementation success or objective correctness. Repeated ratings from an expert and multiple cases from a system must not be treated as independent replications when interpreting the results. Conclusions remain bounded to the sampled requirements, catalog, generated records, and participating experts.

B. Industrial Requirement Sample

The study uses a bounded industrial sample comprising 4 systems, 12 user stories (3 per system), and 35 acceptance criteria (Table II). The supplied data-scope document defines this as a minimal, delimited sample for studying architectural-pattern selection and decision documentation for AI-enabled systems. It is not a representative sample of industrial requirements or AI-enabled systems. The available artifacts do not establish a probabilistic sampling procedure; we therefore do not interpret these counts as population estimates.

Each requirement case consists of the system purpose, one stakeholder user story, and all of that story's acceptance criteria. The source CSV stores one acceptance criterion per row, repeating the corresponding purpose and story. Grouping by the stable user-story identifier reconstructs the case without rewriting the original Portuguese requirement text or changing criterion order. The criterion-row count must not be confused with the number of requirement cases.

System purpose provides contextual information, but is not sufficient on its own to establish pattern applicability. Positive evidence must be grounded in the user story and/or its acceptance criteria. The model is not authorized to infer unstated technologies, requirements, or constraints from the purpose.

The source uses generic names `System_01`–`System_04`; the paper uses the corresponding stable identifiers `SYS01`–`SYS04`. Story and criterion identifiers preserve traceability within each system without exposing real system names. The organization is not named in this review draft. According to the supplied scope document, the study input is restricted to purposes, identifiers, user stories, and acceptance criteria; it does not include operational databases, personal data, credentials, customer records, or production content. This delimitation does not itself establish participant consent, ethics approval, or permission for unrestricted public release.

TABLE I
RESEARCH QUESTIONS, UNITS OF ANALYSIS, AND EVIDENCE MAPPING.

ID	Research question	Unit, evidence, and intended analysis
RQ1	How does the LLM classify architectural-pattern applicability across the requirement cases, and which patterns does it identify as applicable?	Unit: one user-story–pattern pair. Evidence: the official exhaustive assessment log, original requirements, and pattern catalog. Analysis: distributions of the three classifications and applicable-pattern sets by story and system, retaining zero-positive stories.
RQ2	To what extent do the patterns classified as applicable by the LLM overlap with the frozen RAG baseline's final Top-5 candidates for the same requirement cases?	Unit: one LLM-positive user-story–pattern pair, grouped by story. Evidence: official applicability assessments and the frozen final Top-5 rankings. Analysis: membership-intersection counts by story and overall, with the LLM-positive set as an explicit denominator; no applicability-accuracy metric.
RQ3	How do experts assess the applicability of the LLM-selected patterns and the adequacy, grounding, consequences, and perceived usefulness of the corresponding generated ADRs?	Unit: one expert's rating of one selected pair and its ADR on one questionnaire item. Evidence: consolidated expert responses to Q1–Q5 and optional comments for the prepared cases. Analysis: item-specific ordinal distributions and case-level summaries with response counts; comments contextualize ratings. Aggregation and inclusion rules must be specified before consolidated analysis.

TABLE II
THE BOUNDED INDUSTRIAL REQUIREMENT SAMPLE.

System ID	User stories	Acceptance criteria
SYS01	3	6
SYS02	3	11
SYS03	3	7
SYS04	3	11
Total	12	35

C. Architectural Pattern Catalog

We use 70 pattern records from the repository associated with Heiland et al.'s multivocal literature review of design patterns for AI-based systems [1], [2]. For provenance verification, we compared the local files with the public repository at revision `9ff3e61` (the full revision is recorded in the artifact audit). All 70 records matched the corresponding repository files byte-for-byte, as checked using Git blob hashes. This identifies the source snapshot of the catalog; it does not establish the effectiveness of the individual patterns.

Each pattern is stored as one JSON record with a stable three-digit identifier, from `001` through `070`. The applicability prompt supplies exactly five catalog fields: *ID*, identifying the record; *name*, naming the pattern; *motivation*, describing the problem or need; *solution*, describing its proposed response; and *consequences*, describing resulting effects and trade-offs. Identifiers remain strings, preserving leading zeros and enabling joins with the assessment and ADR artifacts. Source text is preserved rather than normalized or rewritten.

The records also contain aliases, examples, related-pattern identifiers, categories, and resource identifiers. These additional fields are not supplied to the applicability model. Resource identifiers provide source pointers within the repository, not independently verified bibliographic descriptions in the local catalog. We therefore attribute the collection to the verified repository and its associated publication rather than infer individual source citations from numeric resource codes.

All five input fields are present as strings in every record. Explicitly empty motivation or consequences strings are retained as empty inputs, without inventing replacement descriptions. A missing field or non-string value is an input-validation failure, not an invitation to repair the source. This preservation policy keeps the assessed catalog traceable while acknowledging uneven descriptive completeness.

The catalog includes general architectural patterns, such as Client–Server, alongside patterns concerning machine-learning safeguards and model serving. We use the records as a bounded candidate set, not as equally mature, mutually independent, or universally applicable solutions. Catalog inclusion alone does not establish applicability to a requirement case; that relationship is assessed separately.

REFERENCES

[1] L. Heiland, M. Hauser, and J. Bogner, "Design patterns for AI-based systems: A multivocal literature review and pattern repository," in *2023 IEEE/ACM 2nd International Conference on AI Engineering – Software Engineering for AI (CAIN)*. IEEE, 2023, pp. 184–196.

[2] SWE4AI, "Design patterns for AI-based systems: A web-based repository," [Online]. Available: https://swe4ai.github.io/ai-patterns/, 2023, repository snapshot 9ff3e61; accessed October 12, 2026.